EARTH SYSTEM COMPONENTS
12-GEO-M-AG02

Lectures: Miguel D. Mahecha
Exercises: David Montero



UNIVERSITÄT
LEIPZIG

Before Christmas: the atmosphere first, then water

Block	L	Date	Topic
Intro	L01	13.10.2026	Course overview and organisation
I · Atmosphere	L02	20.10.2026	Atmosphere I: concepts and key variables
	L03	**27.10.2026**	**Atmosphere II: radiation balance (today)**
	L04	03.11.2026	Atmosphere III: structure of the atmosphere
	L05	10.11.2026	Atmosphere IV: buoyancy forces
	L06	17.11.2026	Atmosphere V: circulation
II · Terrestrial hydrology	L07	24.11.2026	Terrestrial hydrology I: water-cycle concepts
	L08	01.12.2026	Terrestrial hydrology II
	L09	08.12.2026	Terrestrial hydrology II: measurements, Budyko framework
III · Biogeochemistry	L10	15.12.2026	Biogeochemistry I: element cycles – the carbon cycle

After Christmas: carbon, classification and special lectures

Block	L	Date	Topic
–	–	*22.12.–02.01.*	*Christmas break*
III · Biogeochemistry	L11	05.01.2027	Biogeochemistry II: terrestrial biogeochemistry
IV · Classification & special topics	L12	12.01.2027	Climate classification systems and their limitations
	L13	19.01.2027	Atmospheric rivers
	L14	26.01.2027	Forest damage and climate change – guest lecture: Prof. Henrik Hartmann
	L15	02.02.2027	Climate classifications and vegetation
–	–	*Feb 2027*	*Oral exams (dates tbc)*

RECAP … REMEMBER

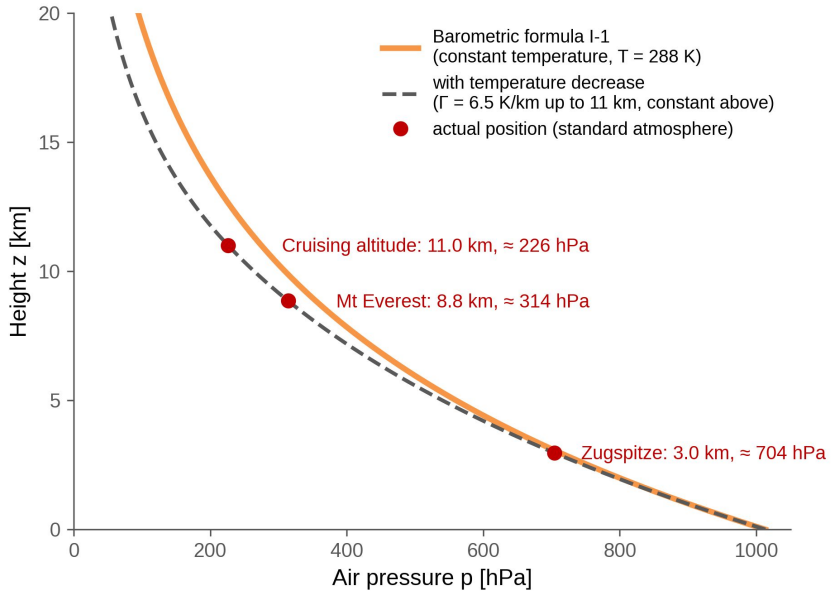
Pressure in the atmosphere decreases with height …

Formula sheet I-1

$$p(z) = p(z_0)\,\exp\left[-\frac{M_\mathrm{L}\,g}{R\,T}\,(z - z_0)\right]$$

$p(z)$, $p(z_0)$	air pressure at height z and at reference height z_0 [Pa]
z, z_0	height, reference height [m]
M_L	molar mass of dry air, $0.028\,97\,[\mathrm{kg\,mol^{-1}}]$
g	gravitational acceleration at the Earth's surface, $9.81\,[\mathrm{m\,s^{-2}}]$
R	universal gas constant, $8.314\,[\mathrm{J\,mol^{-1}\,K^{-1}}]$
T	temperature, assumed constant [K]

Assumptions: hydrostatic equilibrium, constant temperature, constant g and M_L.

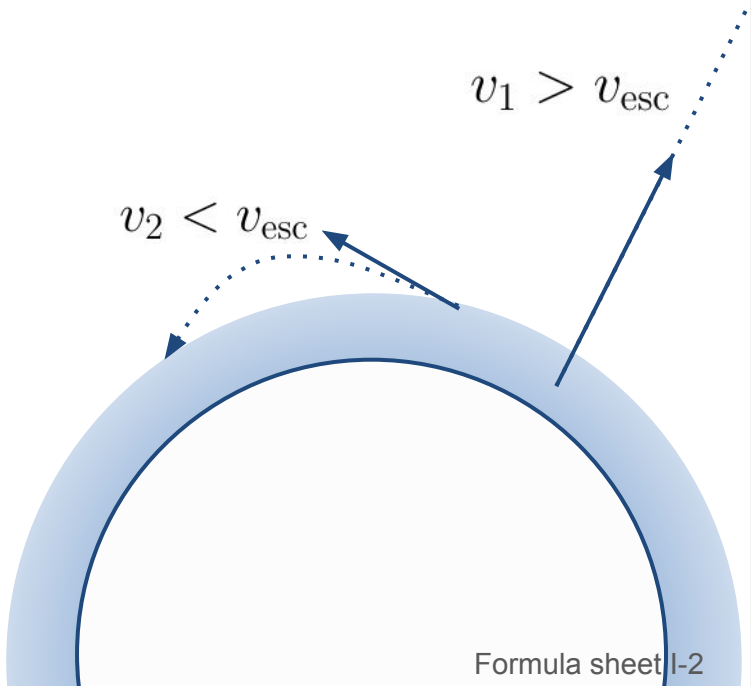


The barometric formula describes the decrease of pressure with height in an atmosphere (assuming constant temperature)

Why can a planet retain an atmosphere?

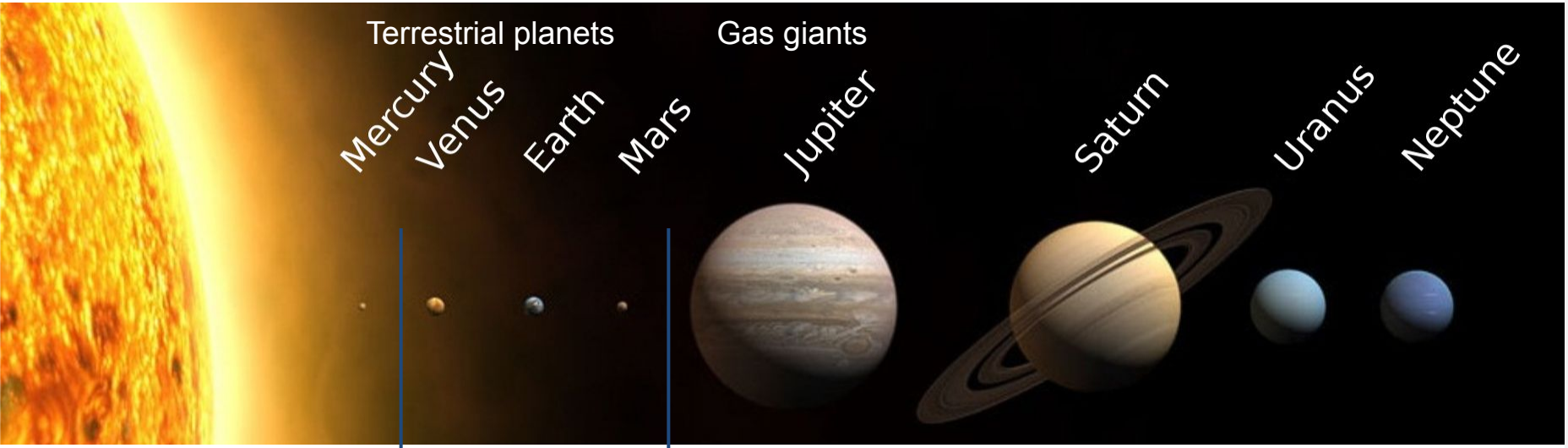
Every planet has a characteristic escape velocity v_{esc}, i.e. the speed needed to leave its gravity field:

What are the determinants that allow a planet to keep its atmosphere?



Formula sheet I-2

What do we know about different atmospheres?



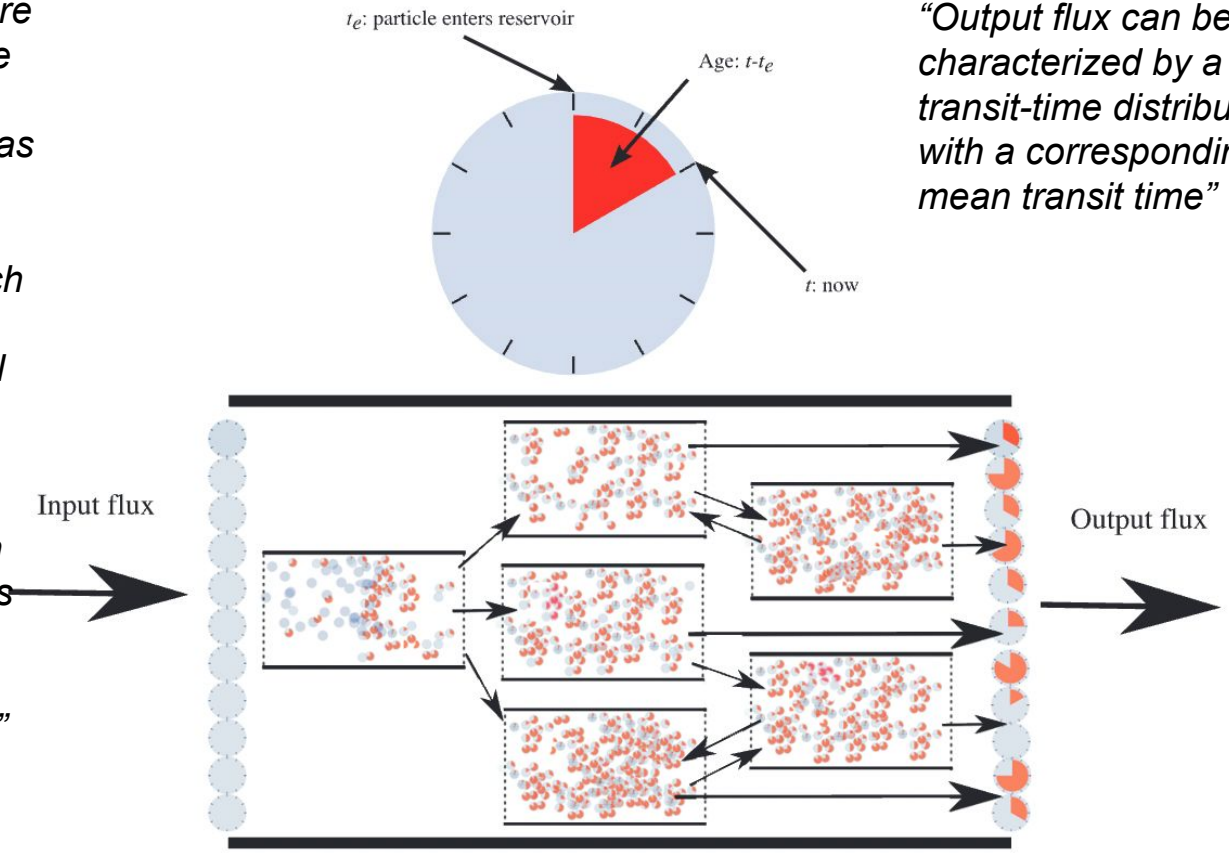
Both light and heavy gases are lost	Light gases are lost	Everything can be retained
Solid surface	Solid surface	No solid surface

How you should really think about this

Sierra, C. A., et al. (2017). *Global Change Biology*, *23*(5), 1763-1773

"Each molecule is represented here as a clock that measures their age in the system. A system (e.g. the atmosphere) can be represented as a set of pools (rectangles) with mass transfer among each other (arrows), with an input flux in which particles enter with age=0. At any given time t, particles in each pool have different ages and therefore each pool has a pool-age distribution with a corresponding mean pool age. For all particles in the system at any given time t, it is also possible to define a system-age distribution with a corresponding mean system age." Sierra et al. (2017)

(Text slightly adapted)



"Output flux can be characterized by a transit-time distributio with a corresponding mean transit time"

WHAT IS THE AVERAGE TEMPERATURE OF THE EARTH?

Questions we need to ask:

1. How much radiant energy does the Sun emit?
2. How much of it reaches the Earth?
3. How much total radiation does the Earth absorb?
4. How much does the Earth radiate in return?
5. What is the temperature of the Earth?

Wavelength of peak emission is inversely proportional to T

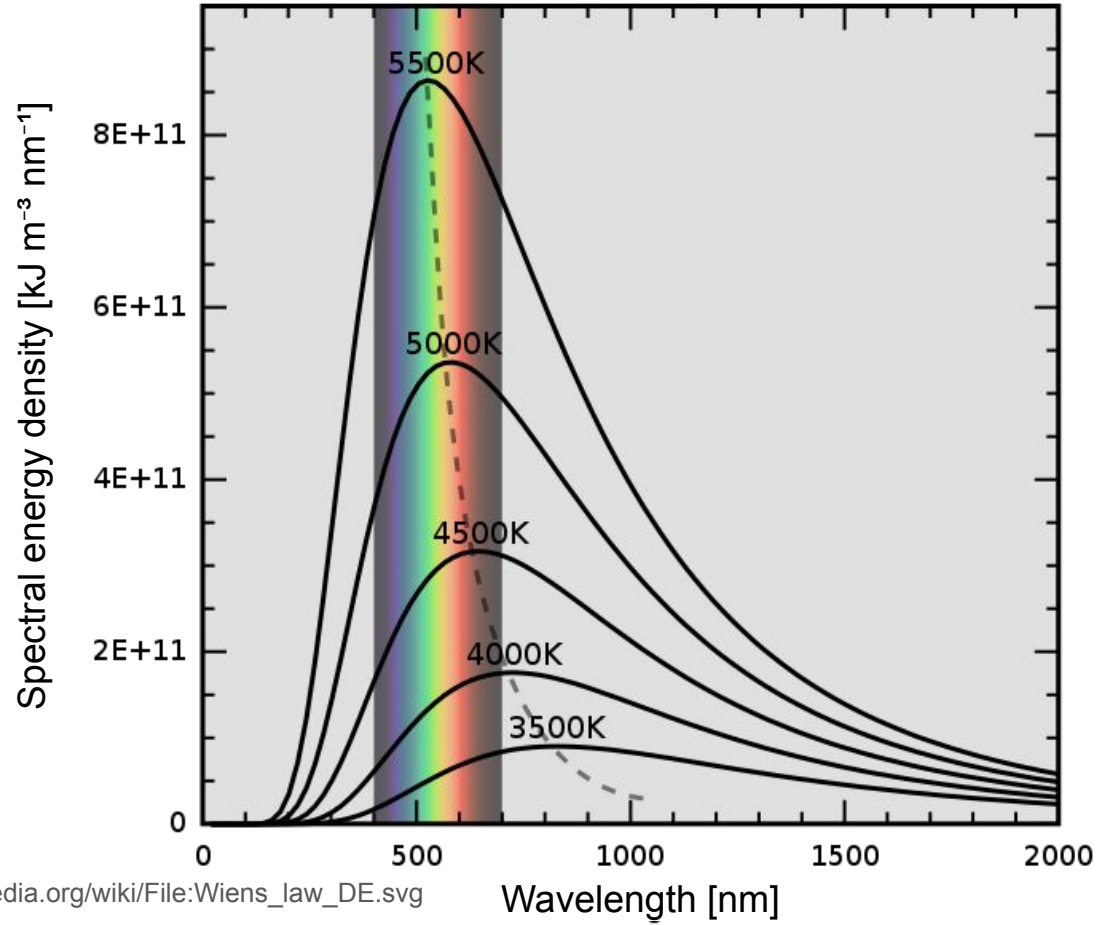
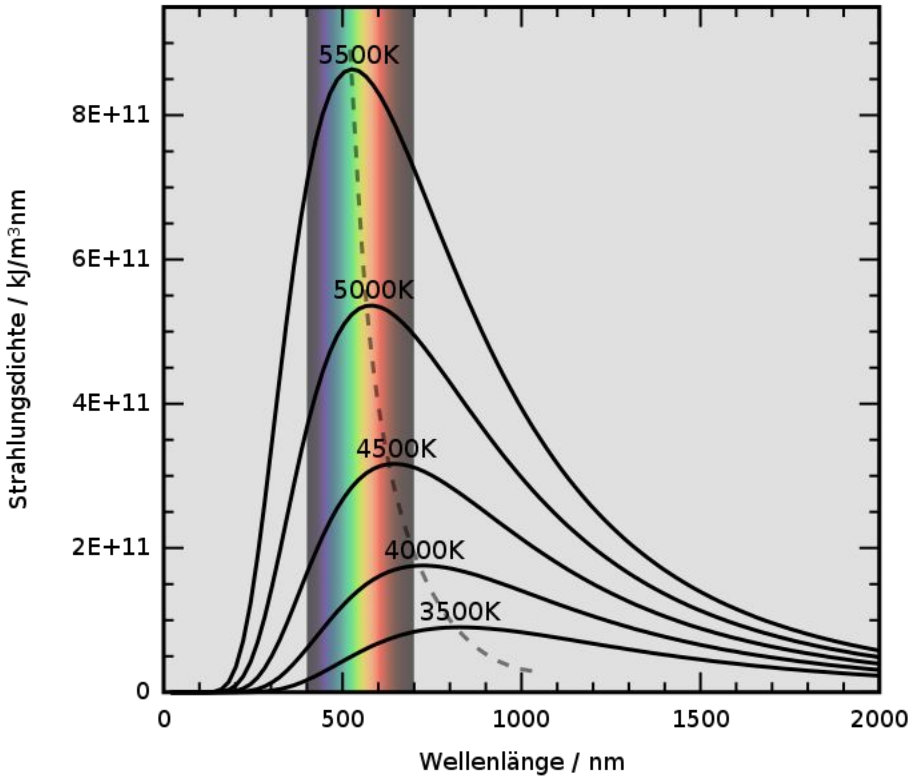


Fig.: https://commons.wikimedia.org/wiki/File:Wiens_law_DE.svg

Formula sheet I-6

Wavelength of peak emission is inversely proportional to T



$$\lambda_{max} = \frac{b}{T}$$

λ_{max}	Wavelength of maximum emission [µm]
b	Wien's constant, 2898 [µm K]
T	Temperature of the emitter [K]

The hotter a body, the shorter the wavelength at which it emits most strongly.

Wien's displacement law states that the wavelength of the most intense radiation is inversely proportional to the temperature of the emitting body.

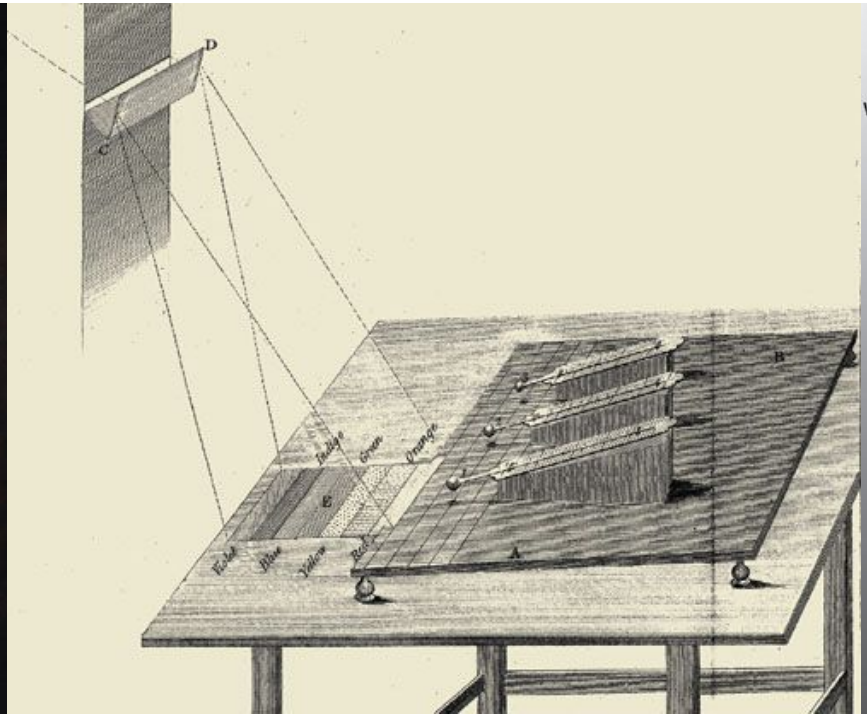
Fig.: https://commons.wikimedia.org/wiki/File:Wiens_law_DE.svg

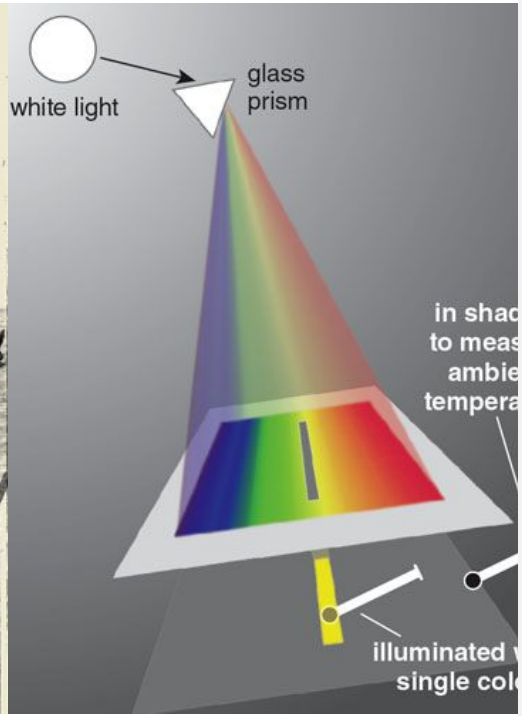
How do we know that there is infrared radiation?

William Herschel, experiments around 1800

https://www.americanscientist.org/article/herschel-and-the-puzzle-of-infrared

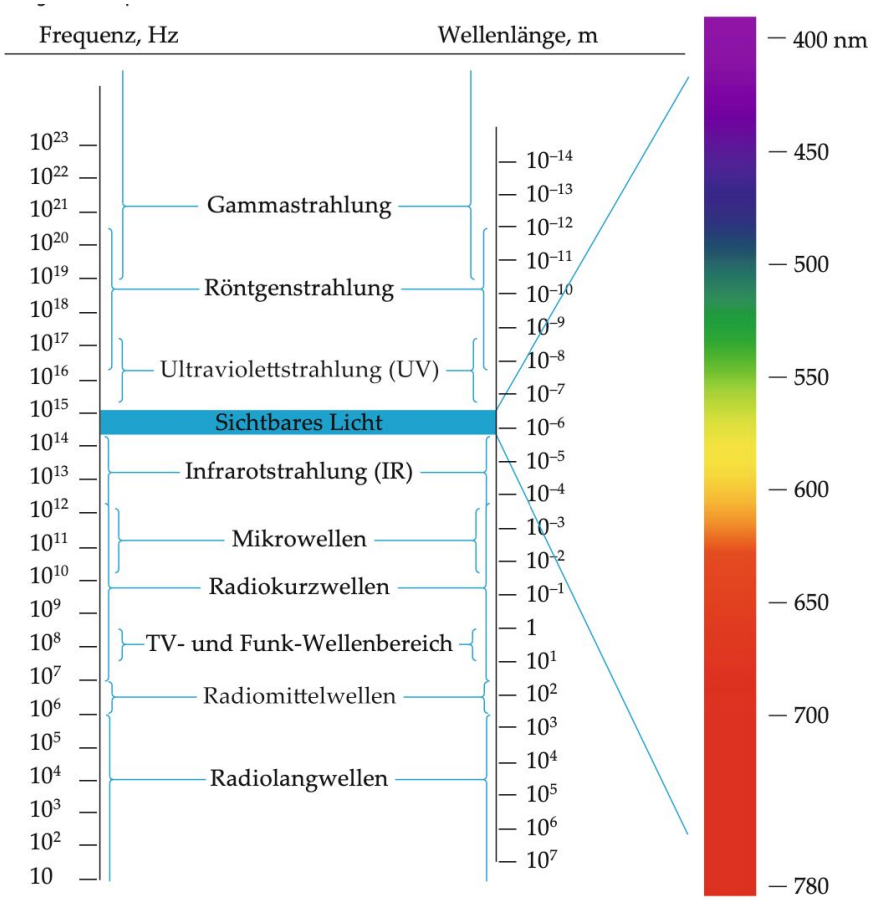






Electromagnetic Radiation

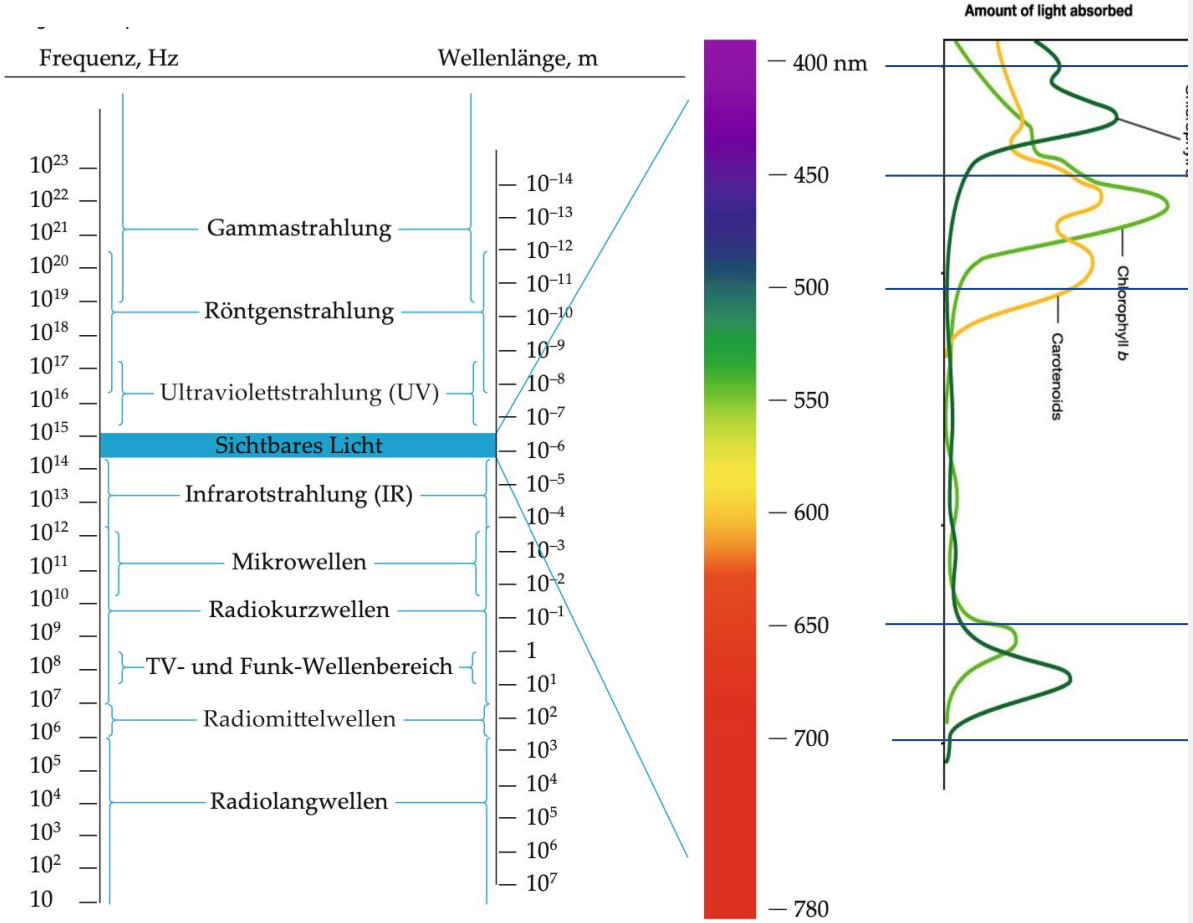
- Only a small part of the electromagnetic spectrum is “visible” to the human eye
- Typical human vision: 380–780 nm



Left: Tipler, P. A., Mosca, G., & Wagner, J. (2015). Physik für Wissenschaftler und Ingenieure, 2. *J. Wagner*.

Electromagnetic Radiation

- Only a small part of the electromagnetic spectrum is “visible” to the human eye
- Typical human vision: 380–780 nm
- Photosynthetically active radiation (PAR): 400–700 nm



Left: Tipler, P. A., Mosca, G., & Wagner, J. (2015). Physik für Wissenschaftler und Ingenieure, 2. *J. Wagner*.

At which wavelength does the Sun emit most strongly?



$$\lambda_{max} = \frac{b}{T}$$

λ_{max}	Wavelength of maximum emission [µm]
b	Wien's constant, 2898 [µm K]
T	Temperature of the emitter [K]

The hotter a body, the shorter the wavelength at which it emits most strongly.

Fig.: https://photomeeting.de/galleries/bestof/content/20040608a_Kopie_large.html

Formula sheet I-6

Light of the Sun

Colour temperature of the Sun

$$\text{Sun:} \quad \lambda_{max} = \frac{2898\,[\mu m\,K]}{5778\,[K]} \approx 0{,}50\,[\mu m] \quad \Rightarrow \text{ visible ligh}$$

λ_{max}	Wavelength of maximum emission [µm]
T	Surface temperature: Sun ≈ 5778 [K], Earth ≈ 288 [K]

Sun: shortwave radiation (light). Earth: longwave thermal radiation.

Light of the Sun

$$\text{Sun:} \quad \lambda_{\max} = \frac{2898\,[\mu\text{m K}]}{5778\,[\text{K}]} \approx 0{,}50\,[\mu\text{m}] \quad \Rightarrow \text{ visible ligh}$$

$$\text{Earth:} \quad \lambda_{\max} = \frac{2898\,[\mu\text{m K}]}{288\,[\text{K}]} \approx 10\,[\mu\text{m}] \quad \Rightarrow \text{ Infrared}$$

Why do we have this temperature on Earth?

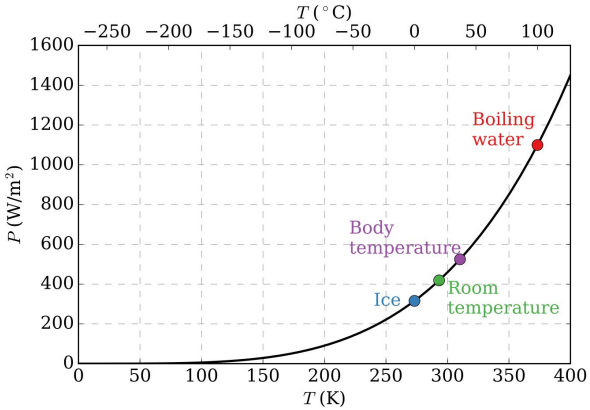
$\lambda_{\max}$	Wavelength of maximum emission [µm]
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Sun: shortwave radiation (light). Earth: longwave thermal radiation.

Some physical principles: the Stefan–Boltzmann law

$$I = \varepsilon \sigma T^4$$

I	Radiant flux density (emitted power per unit area) $[\mathrm{W\,m^{-2}}]$
ε	Emissivity, $0 \leq \varepsilon \leq 1$ (Black body: 1) $[-]$
σ	Stefan-Boltzmann constant, $5{,}670 \cdot 10^{-8}\,[\mathrm{W\,m^{-2}\,K^{-4}}]$
T	Temperature [K]



The “T to the power of 4” law (colloquially) states that the thermally radiated power of an ideal black body is determined solely by its temperature. → All bodies with a temperature > 0 K emit thermal radiation!

What happens if I double the (absolute) temperature of an object?

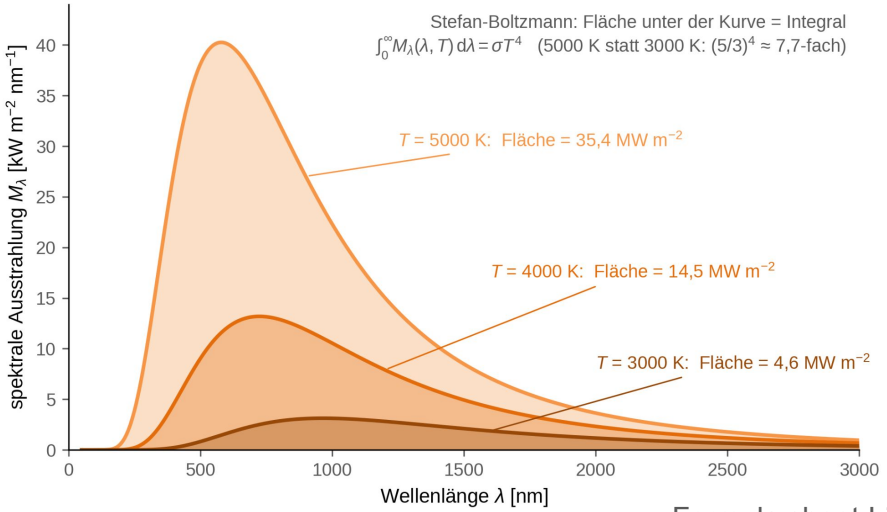
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T	Temperature [K]

Radiation is “electromagnetic wave energy”, expressed as “radiant flux density” (W m⁻²).

The Stefan-Boltzmann law describes the radiant power integrated over all frequencies/wavelengths.



Formula sheet I-7

Let's put some things together

What is the easiest way to estimate the temperature of the Sun?

Let's put some things together

- We can measure the wavelength of maximum emission of the Sun, e.g. with a spectrometer
- Inserting this value lets us estimate the temperature of the Sun

$$\lambda_{\max} = \frac{b}{T} \quad \Big| \cdot T \quad \Big| : \lambda_{\max}$$

$$T = \frac{b}{\lambda_{\max}}$$

$$T_\odot = \frac{2898\,[\mu\mathrm{m\,K}]}{0{,}5\,[\mu\mathrm{m}]} \approx 5800\,[\mathrm{K}]$$

$\lambda_{\max}$	measured wavelength of maximum solar emission, $\approx 0{,}5\,[\mu\mathrm{m}]$ (green-blue light)
b	Wien's constant, $2898\,[\mu\mathrm{m\,K}]$
$T_\odot$	Surface temperature of the Sun [K]

µm cancels out, leaving K. The "colour" of sunlight gives the temperature of the Sun's surface (precise value: $\approx 5778\,[\mathrm{K}]$).

Formula sheet I-6

Let's put some things together

How much does the sun emit?

Or, more formally what is the radiant flux density of the Sun?

Let's put some things together

- We can measure the wavelength of maximum emission of the Sun, e.g. with a spectrometer
- Inserting this value lets us estimate the temperature of the Sun: 5764.5 K
- Using the Stefan–Boltzmann law, we get …

$$
\begin{aligned}
I_\odot &= \sigma T_\odot^4 \\
&= 5{,}670 \cdot 10^{-8}\,[\mathrm{W\,m^{-2}\,K^{-4}}] \cdot (5764{,}5\,[\mathrm{K}])^4 \\
&\approx 6{,}26 \cdot 10^{7}\,[\mathrm{W\,m^{-2}}] = 62\,600\,[\mathrm{kW\,m^{-2}}]
\end{aligned}
$$

$I_\odot$	Radiant flux density at the Sun's surface $[\mathrm{W\,m^{-2}}]$
σ	Stefan-Boltzmann constant, $5{,}670 \cdot 10^{-8}\,[\mathrm{W\,m^{-2}\,K^{-4}}]$
$T_\odot$	Surface temperature of the Sun (from Wien), $\approx 5764{,}5\,[\mathrm{K}]$

Units: $[\mathrm{K^{-4}}] \cdot [\mathrm{K^4}]$ *cancels out, leaving* $[\mathrm{W\,m^{-2}}]$. *Each square metre of the Sun's surface emits around* $63\,\mathrm{MW}$.

Formula sheet I-6, I-7

Let’s put some things together

But how do we get the total emission of the Sun?

Let's put some things together

- We can measure the wavelength of maximum emission of the Sun, e.g. with a spectrometer
- Inserting this value lets us estimate the temperature of the Sun: 5764.5 K
- Using the Stefan–Boltzmann law, we get … ≈ 62 600 kW m⁻² (6.26 × 10⁷ W m⁻²)
- Knowing the surface area of the Sun (via its radius), we get the total emission of the Su

$$\begin{aligned}\Phi_\odot &= I_\odot \cdot 4\pi r_\odot^2 \\ &= 6{,}26 \cdot 10^7\,[\mathrm{W\,m^{-2}}] \cdot 4\pi \cdot \left(6{,}96 \cdot 10^8\,[\mathrm{m}]\right)^2 \\ &\approx 3{,}81 \cdot 10^{26}\,[\mathrm{W}]\end{aligned}$$

$\Phi_\odot$	Radiant power of the entire Sun [W]
$I_\odot$	Radiant flux density at the Sun's surface, $\approx 6{,}26 \cdot 10^7\,[\mathrm{W\,m^{-2}}]$
$r_\odot$	Solar radius, $6{,}96 \cdot 10^8\,[\mathrm{m}]$

Emission per unit area times the surface area of the sphere $4\pi r_\odot^2$*;* $[\mathrm{W\,m^{-2}}] \cdot [\mathrm{m^2}] = [\mathrm{W}]$*. Combined:* $\Phi_\odot = 4\pi r_\odot^2\,\sigma\,T_\odot^4$.

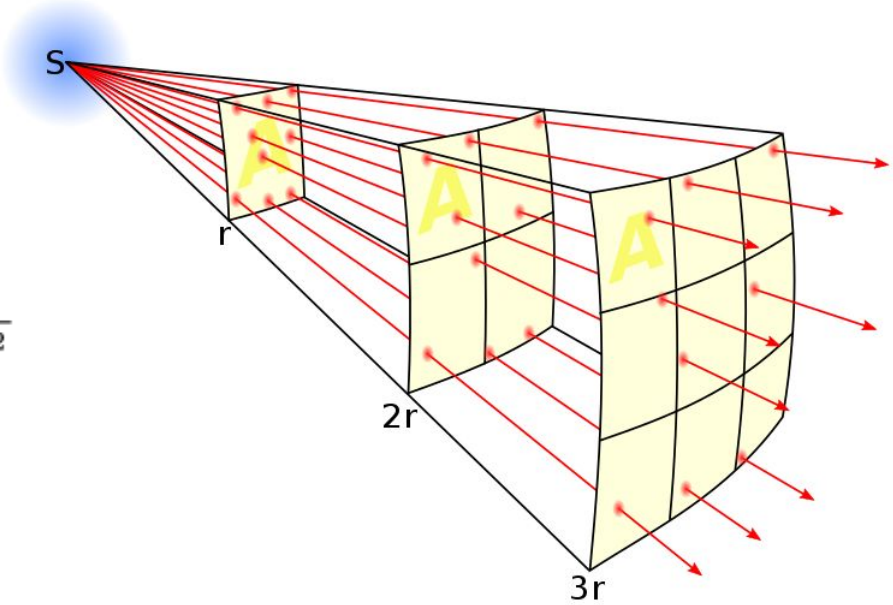
Formula sheet I-6, I-7, I-8

Questions we need to ask:

1. How much radiant energy does the Sun emit?
2. How much of it reaches the Earth?
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5. What is the temperature of the Earth?

How much of the emitted radiation reaches the Earth?

$$\text{intensity} \propto \frac{1}{\text{distance}^2}$$



https://en.wikipedia.org/wiki/Inverse-square_law

What do we get from the sun?

$$I_0 = \frac{\Phi_\odot}{4\pi r_{\mathrm{SE}}^2}$$

$$= \frac{3{,}81 \cdot 10^{26}\,[\mathrm{W}]}{4\pi \cdot (1{,}496 \cdot 10^{11}\,[\mathrm{m}])^2}$$

$$\approx 1355\,[\mathrm{W\,m^{-2}}] \quad (\text{measured: } 1361\,[\mathrm{W\,m^{-2}}])$$

The solar constant

= 1361 W m⁻²

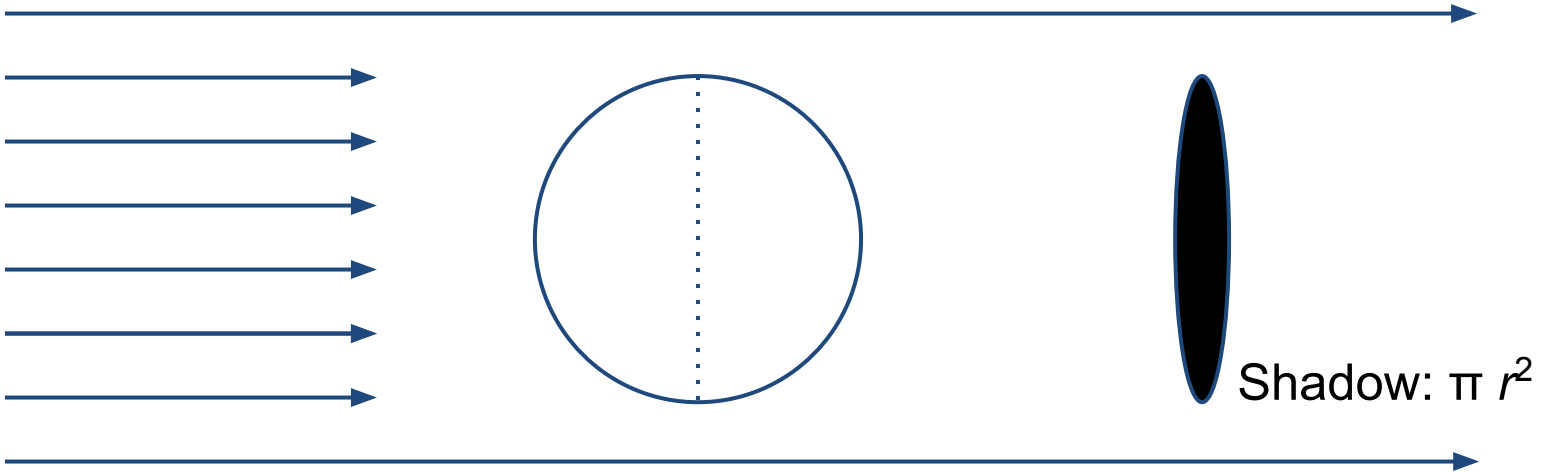
I_0	Solar constant (above the atmosphere, perpendicular incidence)
$\Phi_\odot$	Radiant power of the Sun, $\approx 3{,}81 \cdot 10^{26}$ [W] (Slide 48)
r_{SE}	mean Sun–Earth distance, $1{,}496 \cdot 10^{11}$ [m]

The Sun's power is distributed over a sphere of radius r_{SE}; double the distance ⇒ a quarter of the radiation. The small difference from the satellite value of 1361 comes from estimating $T_\odot \approx 5764{,}5$ [K] using Wien.

Questions we need to ask:

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Solar constant: I_0 = 1361 [W m^{-2}]



$$\Phi_{\text{ein}} = I_0 \cdot \pi r_{\text{E}}^2$$

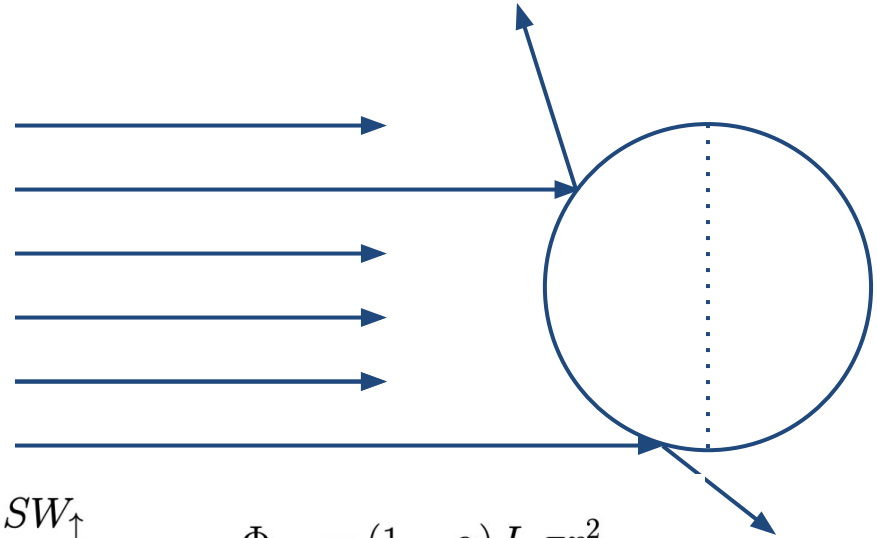
$$= 1361\,[\text{W m}^{-2}] \cdot \pi \cdot \left(6{,}371 \cdot 10^6\,[\text{m}]\right)^2$$

$$\approx 1{,}74 \cdot 10^{17}\,[\text{W}]$$

Φ_{ein}	solar radiation incident on Earth [W]
I_0	Solar constant, 1361 [W m^{-2}]
r_{E}	Radius of Earth, 6,371 · 10^6 [m]

Earth intercepts solar radiation with its projected area (circular disc πr_{E}^2), not its entire surface.

Part of the light gets reflected



Intercepted radiation $I_0 \, \pi r_E^2$

Absorbed radiation $(1 - \alpha) \, I_0 \, \pi r_E^2$

$$\alpha = \frac{SW_\uparrow}{SW_\downarrow} \qquad \Phi_{abs} = (1 - \alpha) \, I_0 \, \pi r_E^2$$

Albedo α = 0 → black body
Albedo α = 1 → perfect mirror

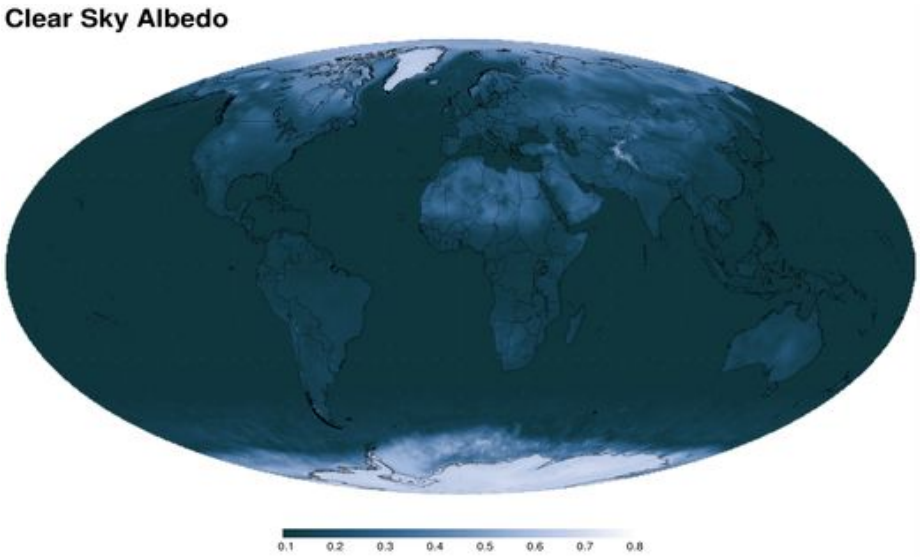
α	Albedo, $0 \leq \alpha \leq 1$ (Earth as a whole: $\approx 0{,}30$) [–]
$SW_\downarrow$, $SW_\uparrow$	incoming, reflected shortwave radiation [$\mathrm{W\,m^{-2}}$]
Φ_{abs}	solar radiation absorbed by Earth [W]
I_0, r_E	Solar constant [$\mathrm{W\,m^{-2}}$], radius of Earth [m]

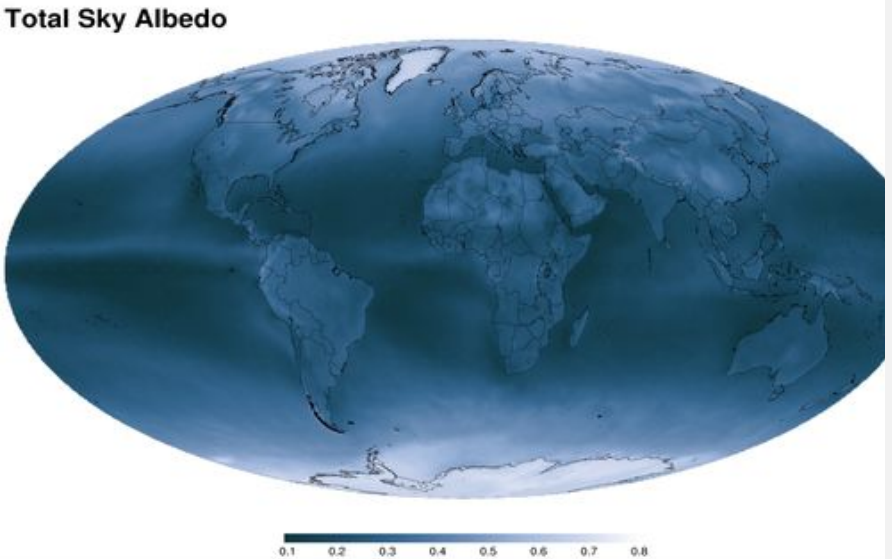
With $\alpha = 0{,}30$: $\Phi_{abs} \approx 0{,}70 \cdot 1{,}74 \cdot 10^{17}$ [W] $\approx 1{,}22 \cdot 10^{17}$ [W]. *Averaged per* m^2 *of Earth's surface:* $(1 - \alpha) \, I_0/4 \approx 238$ [$\mathrm{W\,m^{-2}}$].

Important term: Albedo

Albedo measures the diffuse reflection of radiation.
Albedo is a “fraction” in the range [0; 1].
$\alpha = 0 \rightarrow$ black body; $\alpha = 1 \rightarrow$ perfect mirror.
Usually given for the entire (solar) spectrum

Asphalt:	0.04
Ocean:	0.06
Coniferous forest:	0.08–0.15
Deciduous forest:	0.15–0.1
Desert:	0.4
Concrete:	0.55
Sea ice:	0.5–0.7
Snow:	0.8





Formula sheet I-10

Questions we need to ask:

1. How much radiant energy does the Sun emit?
2. How much of it reaches the Earth?
3. How much total radiation does the Earth absorb?
4. How much does the Earth radiate in return?
5. What is the temperature of the Earth?

The Earth emits radiation proportional to its temperature

Surface of a sphere: $4\pi r^2$

… gives us …

Emission of the Earth: $4\pi r^2 \sigma T^4$

$$A_\mathrm{E} = 4\pi r_\mathrm{E}^2 = 4\pi \cdot \left(6{,}371 \cdot 10^6\,[\mathrm{m}]\right)^2 \approx 5{,}10 \cdot 10^{14}\,[\mathrm{m}^2]$$

$$\Phi_\mathrm{aus} = 4\pi r_\mathrm{E}^2\, \sigma\, T_\mathrm{e}^4$$

A_E	Surface area of the spherical Earth $[\mathrm{m}^2]$
Φ_aus	(longwave) radiation emitted by Earth $[\mathrm{W}]$
σ	Stefan-Boltzmann constant, $5{,}670 \cdot 10^{-8}\,[\mathrm{W\,m^{-2}\,K^{-4}}]$
T_e	Temperature of Earth (without an atmosphere) $[\mathrm{K}]$

Incoming radiation over the projected area πr_E^2, emission over the entire sphere $4\pi r_\mathrm{E}^2$ – hence the factor of 4 later.

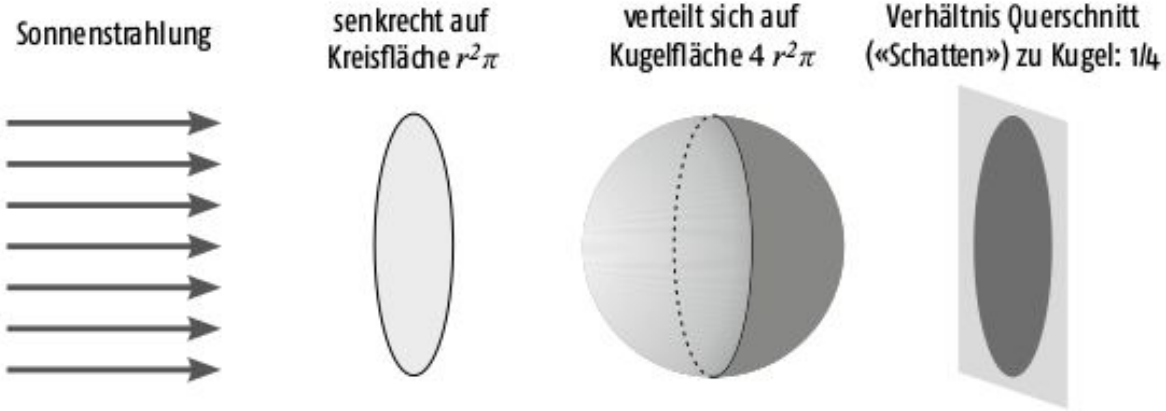


Fig. from Brönnimann (2018) Klimatologie, utb, Haupt Verlag, Bern

Formula sheet I-7, I-11

Questions we need to ask:

1. How much radiant energy does the Sun emit?
2. How much of it reaches the Earth?
3. How much total radiation does the Earth absorb?
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5. What is the temperature of the Earth?

Assuming input equals output …

$$\underbrace{(1-\alpha)\, I_0\, \pi r_{\mathrm{E}}^2}_{\text{absorbed}} = \underbrace{4\pi r_{\mathrm{E}}^2\, \sigma\, T_{\mathrm{e}}^4}_{\text{emitted}}$$

Assuming input equals output …

$$\underbrace{(1-\alpha)\, I_0\, \pi r_{\mathrm{E}}^2}_{\text{absorbed}} = \underbrace{4\pi r_{\mathrm{E}}^2\, \sigma\, T_{\mathrm{e}}^4}_{\text{emitted}}$$

$$(1-\alpha)\, I_0 = 4\, \sigma\, T_{\mathrm{e}}^4$$

Assuming input equals output …

$$\underbrace{(1-\alpha)\, I_0\, \pi r_{\mathrm{E}}^2}_{\text{absorbed}} = \underbrace{4\pi r_{\mathrm{E}}^2\, \sigma\, T_{\mathrm{e}}^4}_{\text{emitted}}$$

$$(1-\alpha)\, I_0 = 4\, \sigma\, T_{\mathrm{e}}^4$$

$$T_{\mathrm{e}}^4 = \frac{(1-\alpha)\, I_0}{4\, \sigma}$$

Assuming input equals output …

$$\underbrace{(1-\alpha)\, I_0\, \pi r_{\mathrm{E}}^2}_{\text{absorbed}} = \underbrace{4\pi r_{\mathrm{E}}^2\, \sigma\, T_{\mathrm{e}}^4}_{\text{emitted}}$$

$$(1-\alpha)\, I_0 = 4\, \sigma\, T_{\mathrm{e}}^4$$

$$T_{\mathrm{e}}^4 = \frac{(1-\alpha)\, I_0}{4\, \sigma}$$

$$T_{\mathrm{e}} = \left(\frac{(1-\alpha)\, I_0}{4\, \sigma} \right)^{1/4}$$

Assuming input equals output …

$$\underbrace{(1-\alpha)\, I_0\, \pi r_\mathrm{E}^2}_{\text{absorbed}} = \underbrace{4\pi r_\mathrm{E}^2\, \sigma\, T_\mathrm{e}^4}_{\text{emitted}}$$

$$(1-\alpha)\, I_0 = 4\,\sigma\, T_\mathrm{e}^4$$

$$T_\mathrm{e}^4 = \frac{(1-\alpha)\, I_0}{4\,\sigma}$$

$$T_\mathrm{e} = \left(\frac{(1-\alpha)\, I_0}{4\,\sigma}\right)^{1/4}$$

$$= \left(\frac{(1-0{,}30)\cdot 1361\,[\mathrm{W\,m^{-2}}]}{4\cdot 5{,}670\cdot 10^{-8}\,[\mathrm{W\,m^{-2}\,K^{-4}}]}\right)^{1/4}$$

Assuming input equals output …

$$\underbrace{(1-\alpha)\, I_0\, \pi r_\mathrm{E}^2}_{\text{absorbed}} = \underbrace{4\pi r_\mathrm{E}^2\, \sigma\, T_\mathrm{e}^4}_{\text{emitted}}$$

$$(1-\alpha)\, I_0 = 4\,\sigma\, T_\mathrm{e}^4$$

$$T_\mathrm{e}^4 = \frac{(1-\alpha)\, I_0}{4\,\sigma}$$

$$T_\mathrm{e} = \left(\frac{(1-\alpha)\, I_0}{4\,\sigma}\right)^{1/4}$$

$$= \left(\frac{(1-0{,}30)\cdot 1361\,[\mathrm{W\,m^{-2}}]}{4\cdot 5{,}670\cdot 10^{-8}\,[\mathrm{W\,m^{-2}\,K^{-4}}]}\right)^{1/4}$$

$$\approx 254{,}6\,[\mathrm{K}] \approx -18\,^\circ\mathrm{C}$$

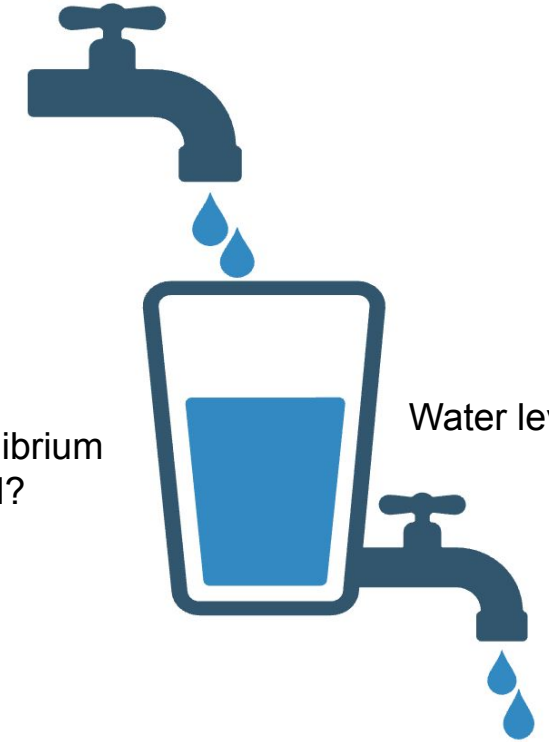
Formula sheet I-11

Questions we need to ask:

1. How much radiant energy does the Sun emit?
2. How much of it reaches the Earth?
3. How much total radiation does the Earth absorb?
4. How much does the Earth radiate in return?
5. What is the temperature of the Earth? … something went wrong!

An analogy for the radiative balance

Input = Solar constant

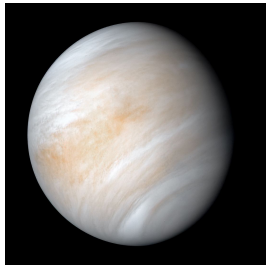


Water level = Temperature

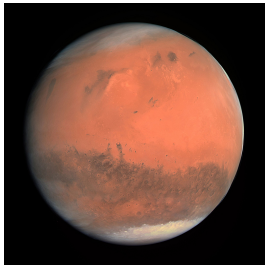
Output = Thermal infrared radiation

- Output is proportional to water level
- + Input → + Water level → + Output → Equilibrium
- What happens when the output gets blocked?

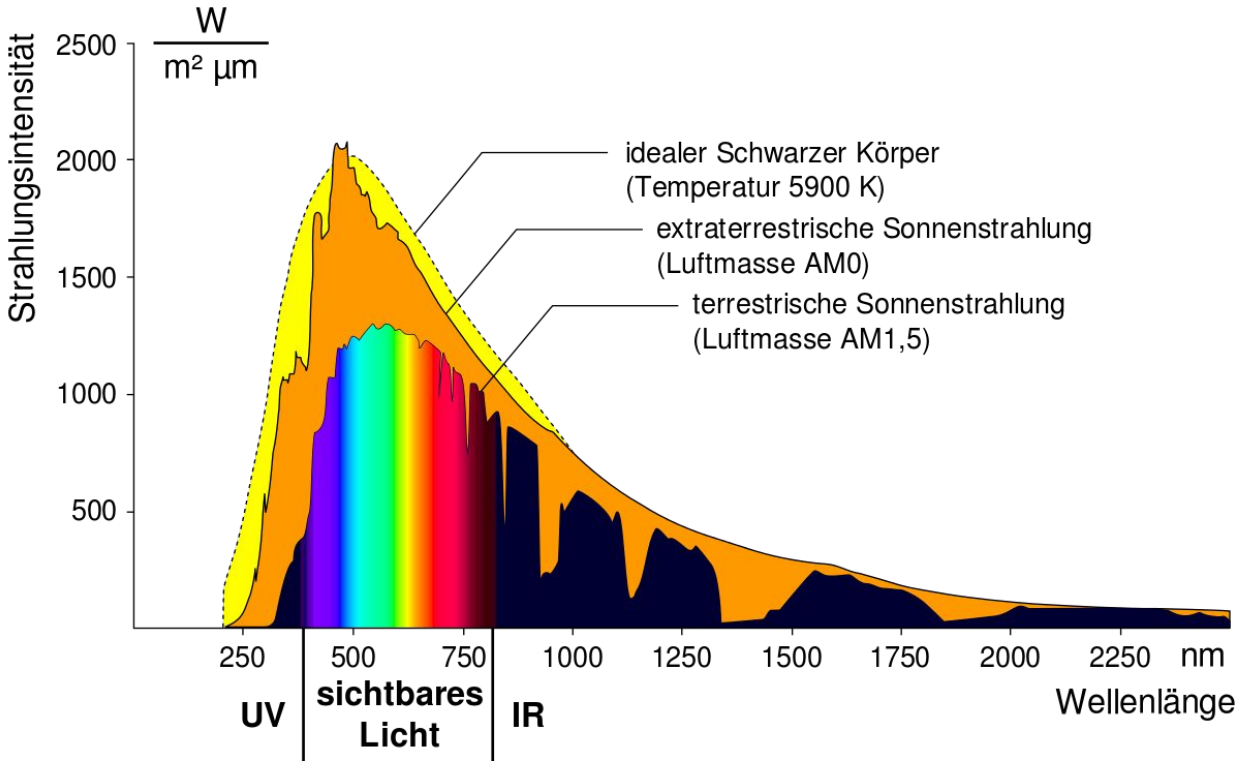
Comparing atmospheres





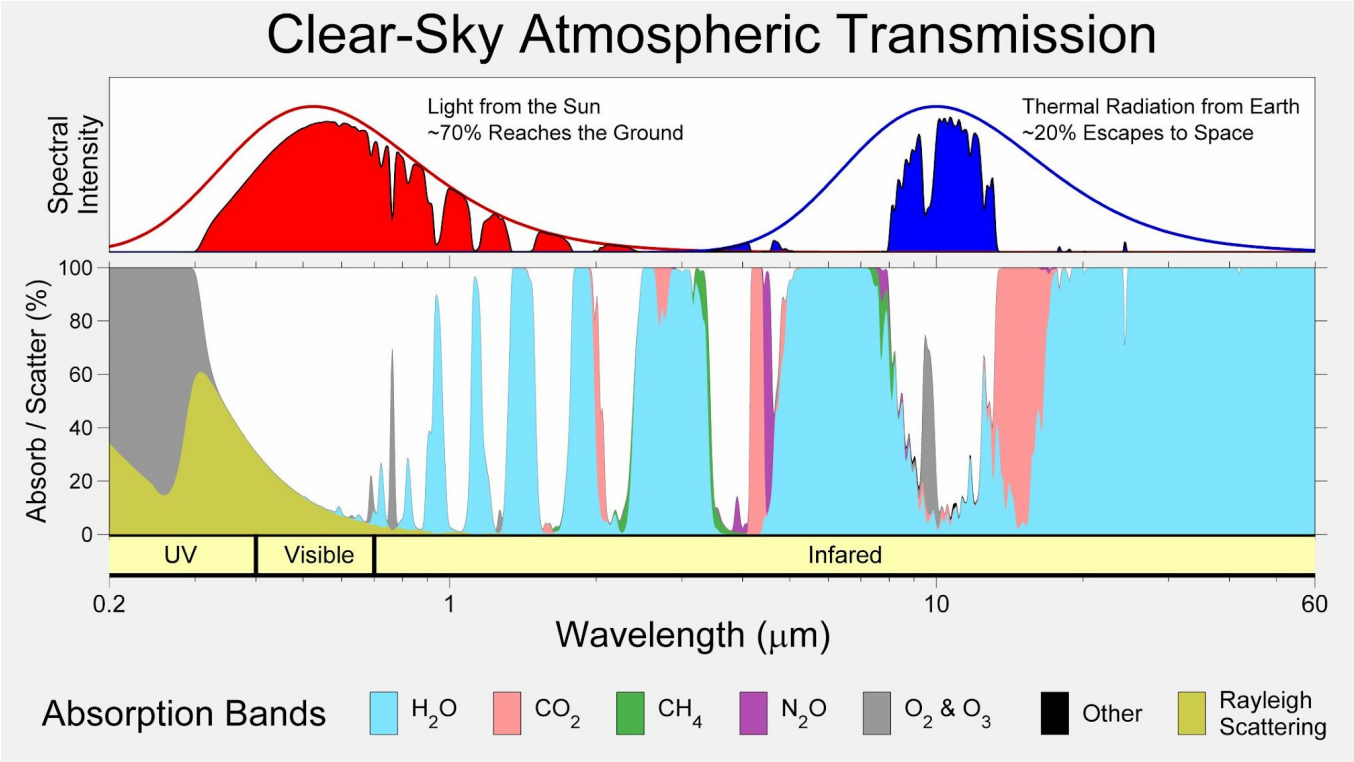


	Venus	Earth	Mars
Albedo	0.71	0.33	0.17
Atmospheric pressure	90 bar	1 bar	0.007 bar
Composition	CO_2 (96 %) N_2 (3.5 %)	N_2 (78 %) O_2 (21 %) Ar (1 %) CO_2 (0.04 %) H_2O (variable)	CO_2 (95 %) N_2 (2.7 %)
Temperature from radiative balance	**-43 °C**	**-21 °C**	**-55 °C**
Observed temperature	**470 °C**	**15 °C**	**-50 °C**
Difference	+513 °C	+36 °C	+5 °C



https://de.wikipedia.org/w/index.php?title=Datei:Sonne_Strahlungsintensitaet.svg&lang=de

Transmissivity of the atmosphere

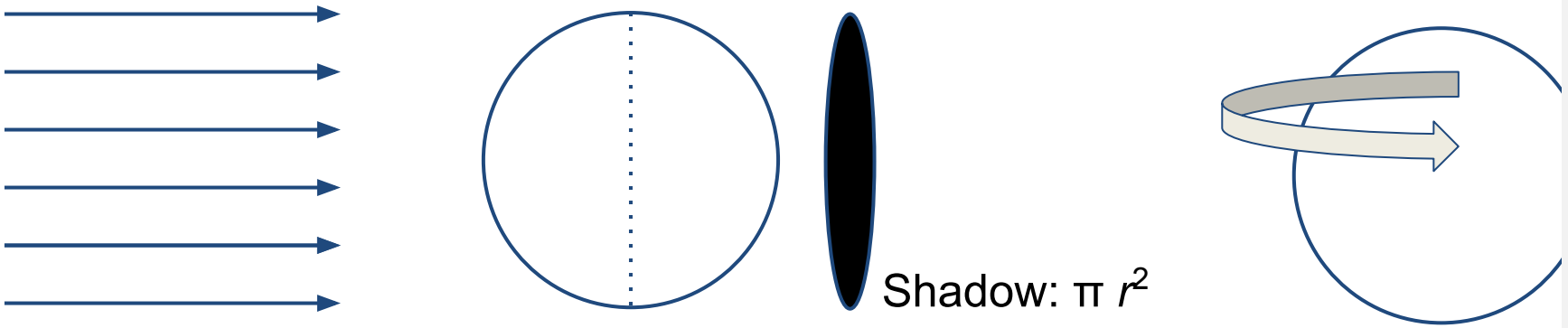


https://twitter.com/RARohde/status/1197147289229385728

How much of the solar constant do we get by surface area?

1361/4 ≈ 340

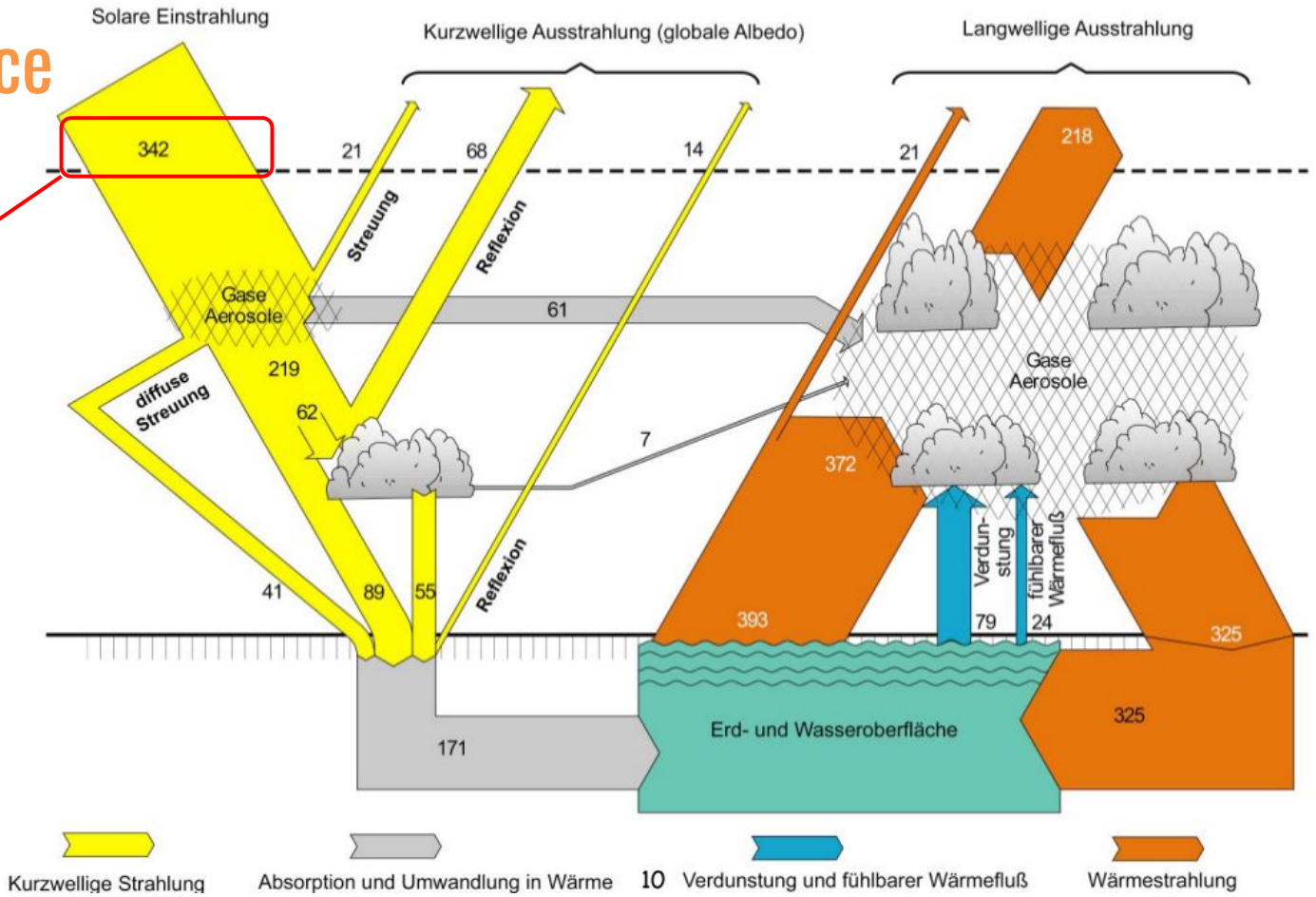
$$\frac{A_1}{A_2} = \frac{\pi r^2}{4\pi r^2} = \frac{1}{4}$$



Formula sheet I-10

Energy balance

We derived this value!



https://www.bgc-jena.mpg.de/bgc-systems/uploads/LecturesAtFSU/VorlesungWS2019_Energiebilanz.key.pdf